The Poisson Distribution

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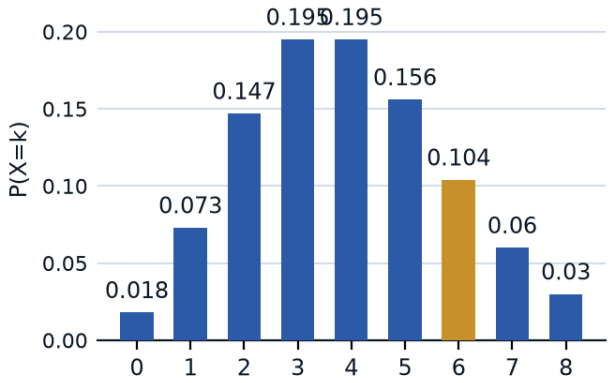
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Conditions for a Poisson distribution**

State the key conditions required for a random variable to be modeled by a Poisson distribution.

2 **[Foundational]** • **Target: Computing Poisson probabilities**

A call center receives an average of 4 calls per minute, modeled by a Poisson distribution with $\lambda = 4$. Find $P(X = 6)$, using $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$, to three decimal places.



3 **[Foundational]** • **Target: Computing Poisson probabilities**

Using the same call center ($\lambda = 4$ calls per minute), find the probability of receiving at most 2 calls in a given minute.

4 **[Foundational]** • **Target: Mean and variance of a Poisson distribution**

State the mean and variance of a Poisson distribution with parameter λ, and explain what is distinctive about this relationship compared to other common distributions.

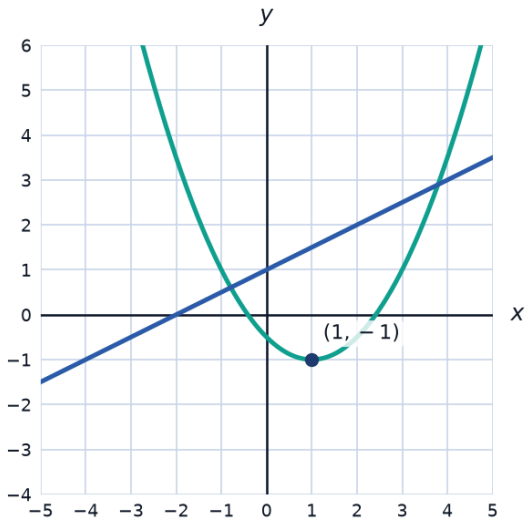
5 **[Foundational]** • **Target: Sums of independent Poisson variables**

Two independent Poisson processes have rates $\lambda_1 = 3$ and $\lambda_2 = 5$ events per hour. If $X_1 \sim \text{Po}(3)$ and $X_2 \sim \text{Po}(5)$, state the distribution of $X_1 + X_2$, and find its mean and variance.

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6 [Foundational] • Target: Poisson as an approximation to the binomial

Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the appropriate value of λ to use.



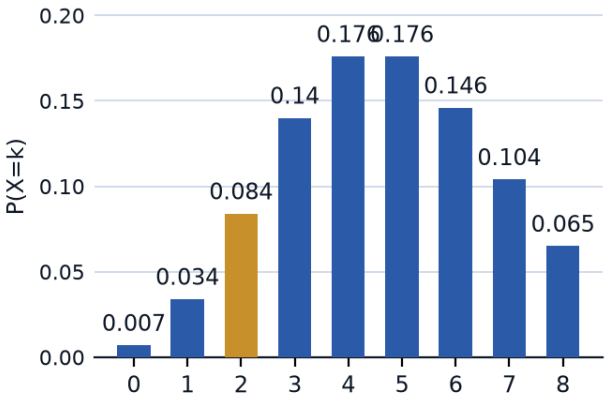
7 [Foundational] • Target: Poisson as an approximation to the binomial

A factory produces 2000 items, each independently defective with probability 0.002. Using the Poisson approximation, find the probability that exactly 3 items are defective, to three decimal places.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Using cumulative Poisson probabilities

A shop receives customers at an average rate of $\lambda = 5$ per 10-minute interval. Find the probability of receiving more than 2 customers in a given 10-minute interval, given $P(X = 0) \approx 0.0067$, $P(X = 1) \approx 0.0337$, and $P(X = 2) \approx 0.0842$.



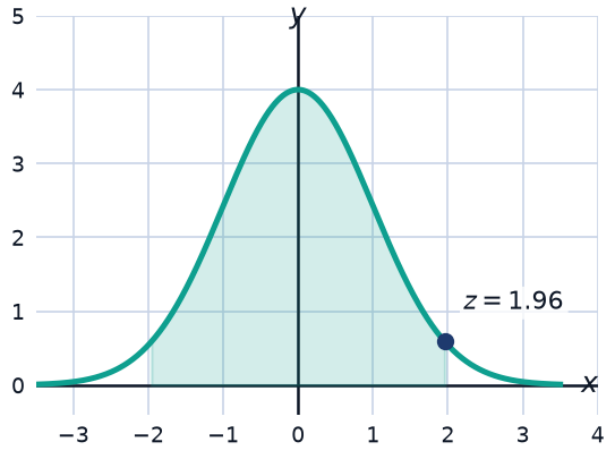
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9 **[Intermediate]** • **Target: Finding an unknown parameter from a given probability**

A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.1353$. Find λ, to the nearest whole number, using $P(X = 0) = e^{-\lambda}$.

10 **[Intermediate]** • **Target: Variance-based reasoning with the Poisson distribution**

A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 3. Find λ, and then find $P(X = \lambda)$, to three decimal places.



11 **[Intermediate]** • **Target: Combining Poisson reasoning with real-world decisions**

A website server crashes on average $\lambda = 0.5$ times per week. The IT team wants to know the probability of at least one crash in a given week, given $P(X = 0) \approx 0.6065$. Should the team be concerned if their target is to keep this probability below 30%?

12 **[Intermediate]** • **Target: Combining a Poisson approximation with conditional probability**

Machine faults occur at an average rate of $\lambda = 1.5$ per day. Find the probability that exactly one fault occurs today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.2231$ and $P(X = 1) \approx 0.3347$.

13 **[Intermediate]** • **Target: Combining Poisson processes over different time intervals**

A radioactive source emits particles at a rate of $\lambda = 2$ per second, modeled by a Poisson distribution. Find $P(X = 3)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.

14 **[Intermediate]** • **Target: Combining Poisson processes over different time intervals**

Explain why the Poisson distribution is not an appropriate model for the number of heads obtained when flipping a fair coin 10 times, referencing the conditions required for a Poisson model.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

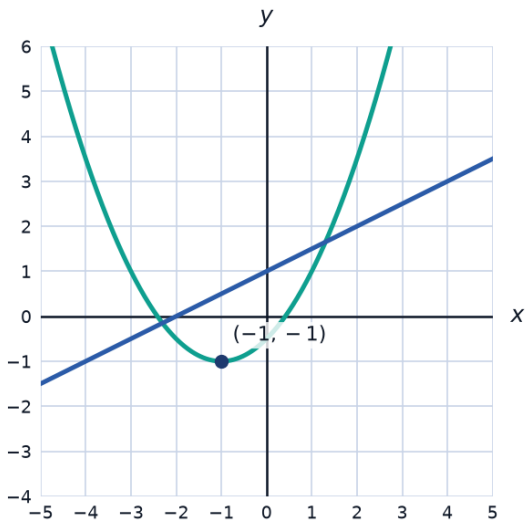
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15 [Challenging] • Target: Comparing two independent Poisson processes • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Two independent Poisson processes have rates $\lambda_A = 3$ and $\lambda_B = 2$ per hour. Find the probability that, in a given hour, process A produces exactly 2 events and process B produces exactly 1 event, to three decimal places, given $P(X_A = 2) \approx 0.2240$ and $P(X_B = 1) \approx 0.2707$.

16 [Challenging] • Target: Rescaling the Poisson parameter for a different interval • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

A radio station receives an average of 12 requests per hour. Find the appropriate Poisson parameter λ to use for a 15-minute segment.



17 [Challenging] • Target: The mode of a Poisson distribution • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

A random variable $X \sim \text{Po}(4)$ has an integer parameter $\lambda = 4$. State the mode(s) of X (the most likely value or values), and explain why a Poisson distribution with an integer λ has two modes, using the ratio $P(X = k)/P(X = k - 1) = \lambda/k$.

18 [Challenging] • Target: Assessing Poisson plausibility from sample data • Cross-Lesson: 5.01 Further Integration Techniques

A biologist counts the number of bacteria colonies in 10 equal-sized petri dishes: $3, 5, 4, 6, 5, 4, 3, 5, 6, 4$. Find the sample mean and the unbiased sample variance, and comment on whether a Poisson model seems plausible for this data.

19 [Challenging] • Target: Changing Time Intervals & Sum of Independent Poisson RVs • Cross-Lesson: 5.02 Differential Equations

If $X \sim B(150, \frac{1}{100})$, use a Poisson approximation to find $P(X = 1)$, correct to 4 decimal places.

20 [Challenging] • Target: Poisson Approximation to the Binomial Distribution • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

If $X \sim B(300, \frac{1}{100})$, use a Poisson approximation to find $P(X = 1)$, correct to 4 decimal places.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 **[Foundational]** • **Target: Identifying an appropriate Poisson scenario**

Which of the following scenarios is best modeled by a Poisson distribution?

a. The number of typing errors on a randomly selected page of a manuscript.

b. The number of heads in 20 coin tosses.

c. The height of a randomly selected adult.

d. The number of correct answers on a 10-question multiple-choice test, each with 4 options.

22 **[Foundational]** • **Target: The Poisson Probability Mass Function**

If $X \sim \text{Po}(\lambda)$, what is the probability mass function $P(X = k)$ for $k = 0, 1, 2, \ldots$?

a. $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$

b. $P(X = k) = \frac{\lambda^k e^{\lambda}}{k!}$

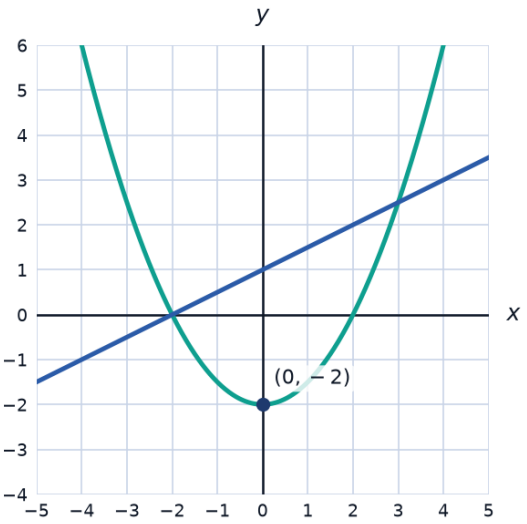
c. $P(X = k) = \binom{n}{k}\lambda^k(1-\lambda)^{n-k}$

d. $P(X = k) = \lambda e^{-\lambda k}$

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23 [Foundational] • Target: The Poisson Probability Mass Function

A call center receives an average of 3 calls per minute ($X \sim \text{Po}(3)$). What is the probability of receiving exactly 2 calls in a given minute?



a. $\frac{9}{2}e^{-3} \approx 0.224$

b. $3e^{-3}$

c. $\frac{27}{6}e^{-3}$

d. e^{-3}

24 [Foundational] • Target: The Poisson Probability Mass Function

For $X \sim \text{Po}(\lambda)$, what is $P(X = 0)$?

a. $e^{-\lambda}$

b. 0

c. 1

d. $\lambda e^{-\lambda}$

25 [Foundational] • Target: Mean, Variance, and Poisson Characteristics

What is a distinctive mathematical property of any Poisson random variable $X \sim \text{Po}(\lambda)$?

a. The mean and variance are strictly equal: $E(X) = \text{Var}(X) = \lambda$

b. The mean is always λ^2

c. The standard deviation equals λ

d. The median always equals $\lambda + 1$

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26 [Foundational] • Target: Mean, Variance, and Poisson Characteristics

If a Poisson random variable has variance $\text{Var}(X) = 16$, what is its standard deviation?

a. 4

b. 16

c. 256

d. $\sqrt{4} = 2$

27 [Foundational] • Target: Mean, Variance, and Poisson Characteristics

Under what practical conditions does a Poisson process serve as a valid model?

a. Events occur independently, singly, and at a constant average rate across continuous time or space

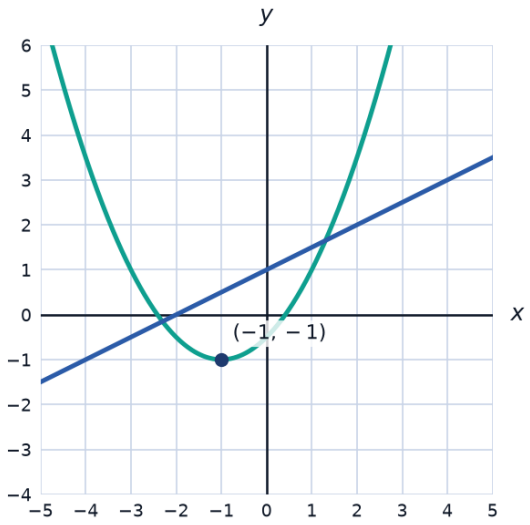
b. There are exactly two trials with equal probability

c. Events must occur in clusters at predetermined times

d. The probability of an event increases as time elapses

28 [Foundational] • Target: Changing Time Intervals & Sum of Independent Poisson RVs

Flaws on a wire occur at an average rate of 1.5 per meter. What is the distribution of flaws Y in a 4-meter segment of wire?



a. $Y \sim \text{Po}(6)$

b. $Y \sim \text{Po}(1.5)$

c. $Y \sim \text{Po}(4)$

d. $Y \sim \text{Po}(9)$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Changing Time Intervals & Sum of Independent Poisson RVs**

If $X \sim \text{Po}(4)$ and $Y \sim \text{Po}(7)$ are independent random variables, what is the distribution of $W = X + Y$?

a. $W \sim \text{Po}(11)$

b. $W \sim \text{Po}(28)$

c. $W \sim \text{Po}(3)$

d. $W \sim N(11, 28)$

30 **[Intermediate]** • **Target: Changing Time Intervals & Sum of Independent Poisson RVs**

If $W \sim \text{Po}(11)$, what is the variance of W?

a. 11

b. 121

c. $\sqrt{11}$

d. 22

31 **[Intermediate]** • **Target: Poisson Approximation to the Binomial Distribution**

When can a Binomial distribution $B(n, p)$ be well approximated by a Poisson distribution $\text{Po}(\lambda)$?

a. When n is very large ($n > 50$) and p is very small ($p < 0.1$), with $\lambda = np$

b. When n is small and $p = 0.5$

c. When $np \geq 10$ and $n(1 - p) \geq 10$

d. For any binomial distribution where $n > 30$

32 **[Intermediate]** • **Target: Poisson Approximation to the Binomial Distribution**

A factory produces microchips with defect probability $p = 0.002$. In a batch of $n = 1000$ chips, approximate the probability of finding zero defects.

a. $e^{-2} \approx 0.135$

b. $e^{-0.002}$

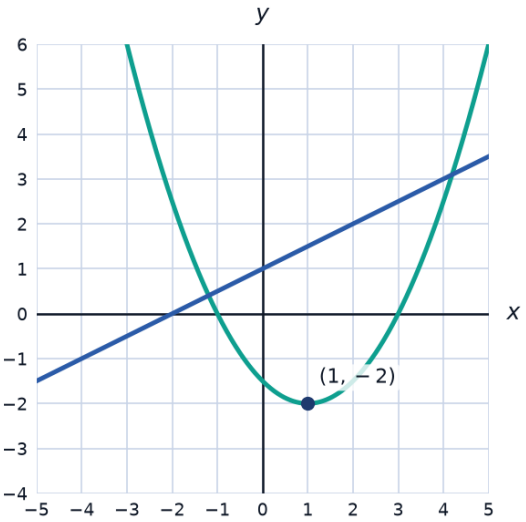
c. $1 - e^{-2}$

d. 0.002

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33 [Intermediate] • Target: Poisson Approximation to the Binomial Distribution

If $X \sim \text{Po}(\lambda)$ has $P(X = 1) = P(X = 2)$, find the value of λ.



a. $\lambda = 2$

b. $\lambda = 1$

c. $\lambda = 0.5$

d. $\lambda = 4$

34 [Intermediate] • Target: The mode of a Poisson distribution

If $X \sim B(300, \frac{1}{100})$, use a Poisson approximation to find $P(X = 3)$, correct to 4 decimal places.

a. 0.2640

b. 0.1680

c. 0.1008

d. 0.2240

35 [Intermediate] • Target: Identifying an appropriate Poisson scenario

If $X \sim B(200, \frac{1}{100})$, use a Poisson approximation to find $P(X = 0)$, correct to 4 decimal places.

a. 0.1753

b. 0.2707

c. 0.1353

d. 0.2153

36 [Intermediate] • Target: Assessing Poisson plausibility from sample data

If $X \sim B(150, \frac{1}{100})$, use a Poisson approximation to find $P(X = 3)$, correct to 4 decimal places.

a. 0.0471

b. 0.1255

c. 0.0141

d. 0.2510

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37 [Intermediate] • Target: The Poisson Probability Mass Function

If $X \sim B(150, \frac{1}{25})$, use a Poisson approximation to find $P(X = 3)$, correct to 4 decimal places.

a. 0.0446

b. 0.0892

c. 0.1606

d. 0.1339

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Mean, Variance, and Poisson Characteristics • Cross-Lesson: 5.01 Further Integration Techniques

If $X \sim B(200, \frac{1}{50})$, use a Poisson approximation to find $P(X = 1)$, correct to 4 decimal places.

a. 0.1954

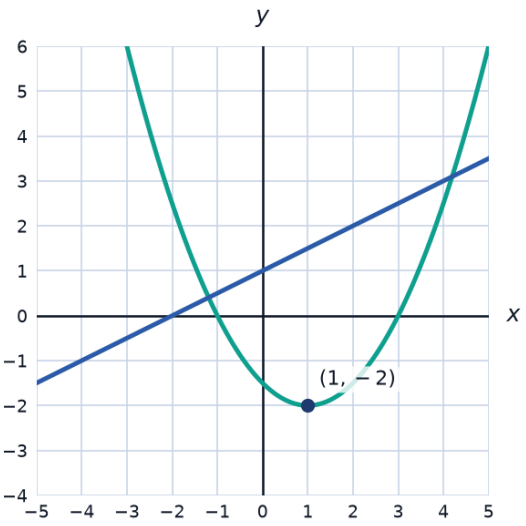
b. 0.0183

c. 0.0733

d. 0.1465

39 [Challenging] • Target: Changing Time Intervals & Sum of Independent Poisson RVs • Cross-Lesson: 5.02 Differential Equations

If $X \sim B(300, \frac{1}{50})$, use a Poisson approximation to find $P(X = 3)$, correct to 4 decimal places.



a. 0.1339

b. 0.0446

c. 0.1606

d. 0.0892

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40 [Challenging] • Target: Poisson Approximation to the Binomial Distribution • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

If $X \sim B(300, \frac{1}{50})$, use a Poisson approximation to find $P(X = 0)$, correct to 4 decimal places.

a. 0.0425

b. 0.0446

c. 0.0025

d. 0.0149

41 [Challenging] • Target: Conditions for a Poisson distribution • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A school offers two optional subjects: Art and Music. The number of students enrolled in each subject and the number of students enrolled in both is shown in the Venn diagram below. The total number of students in the school is 120. Let A be the set of students enrolled in Art and M be the set of students enrolled in Music. Given that $n(A \cup M) = 90$, find the number of students enrolled in neither Art nor Music.

a. 20

b. 30

c. 40

d. 10

42 [Challenging] • Target: Computing Poisson probabilities • Cross-Lesson: 6.01 Matrices (AI HL)

X takes values $0, 1, 2$ with probabilities $\frac{1}{5}, \frac{3}{10}, \frac{1}{2}$. Find $\text{Var}(X)$.

a. $\frac{61}{100}$

b. $\frac{121}{100}$

c. $\frac{73}{100}$

d. $\frac{4}{5}$

43 [Challenging] • Target: Mean and variance of a Poisson distribution • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A discrete random variable X takes values $0, 3, 4$ with probabilities $\frac{1}{10}, \frac{3}{10}, \frac{3}{5}$ respectively. Find $E(X)$.

a. 1

b. 6

c. $\frac{33}{10}$

d. 2

44 [Challenging] • Target: Sums of independent Poisson variables • Cross-Lesson: 6.03 Markov Chains (AI HL)

A discrete random variable X takes values $0, 2, 4$ with probabilities $\frac{1}{10}, \frac{1}{5}, \frac{7}{10}$ respectively. Find $E(X)$.

a. 5

b. 0

c. $\frac{16}{5}$

d. 4

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45 **[Challenging]** • **Target: Poisson as an approximation to the binomial** • **Cross-Lesson: 1.01 Proof by Mathematical Induction**

A discrete random variable X takes values $0, 2, 4$ with probabilities $\frac{3}{10}, \frac{1}{10}, \frac{3}{5}$ respectively. Find $E(X)$.

a. 0

b. 5

c. 1

d. $\frac{13}{5}$

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